

Mathematics Extension 1 Year 11

Inverse Functions

Book 1

Version: 261002

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Syllabus Content

ME1-11-01 solves problems involving inequalities, functions and their inverses, graphical relationships between functions, and parametric equations

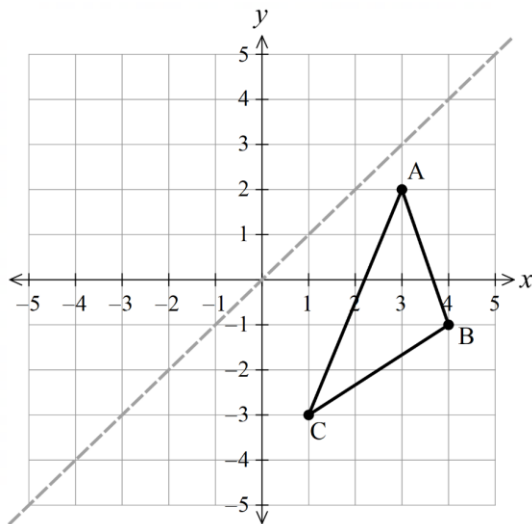
Inverse functions

- Describe a function as one-to-one if every element in the range of the function corresponds to exactly one element of the domain
- Establish that the reflection of a point in the line $y = x$ reverses the coordinates of the point
- Define an inverse function f^{-1} informally as a function that reverses or undoes the effect of the function f
- Recognise that inverse functions exist for one-to-one functions
- Determine the equation for the inverse function $f^{-1}(x)$ of a given one-to-one function $y = f(x)$ by interchanging the variables x and y in $y = f(x)$
- Compare the graphs of a function and its inverse function using graphing applications, and recognise that the two graphs are reflections of each other in the line $y = x$
- Establish that the domain of $f^{-1}(x)$ is the range of $f(x)$, and the range of $f^{-1}(x)$ is the domain of $f(x)$, and use this relationship to solve problems
- Graph the inverse function of a given one-to-one function
- Explain that the reflection in the line $y = x$ exchanges horizontal and vertical lines and recognise that the horizontal line test can therefore be applied to the graph of $y = f(x)$ to determine whether its reflection in the line $y = x$ is a function
- Apply restrictions to the domain of a function, if it is not one-to-one, to obtain an inverse function
- Define formally $f^{-1}(x)$ to be the inverse function of $f(x)$ if the relationships $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$ hold, and use this definition to solve problems
- Solve problems based on the relationship between a function and its inverse function using algebraic and graphical techniques, including determining the points of intersection of a function and its inverse, where they exist

Reflection across $y = x$

🔍 Reflection across $y = x$

Reflect the coordinates A, B, C across the line $y = x$.



Original	A:	B:	C:
Reflection	A':	B':	C':

What do you notice?

📌 Reflecting Across $y = x$

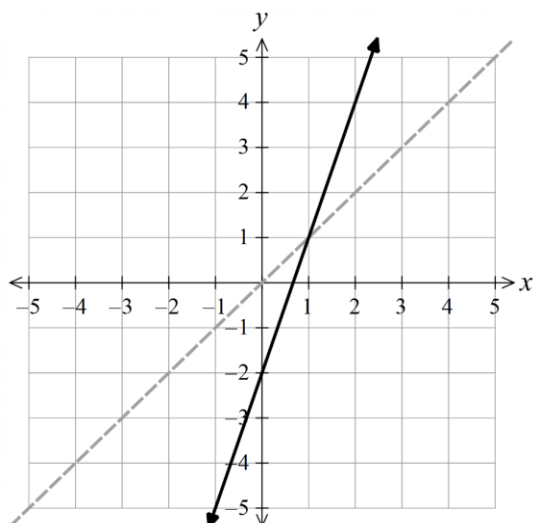
Exchange the x and y -coordinates.

$$\begin{array}{c} (x, y) \\ \downarrow \\ (y, x) \end{array}$$

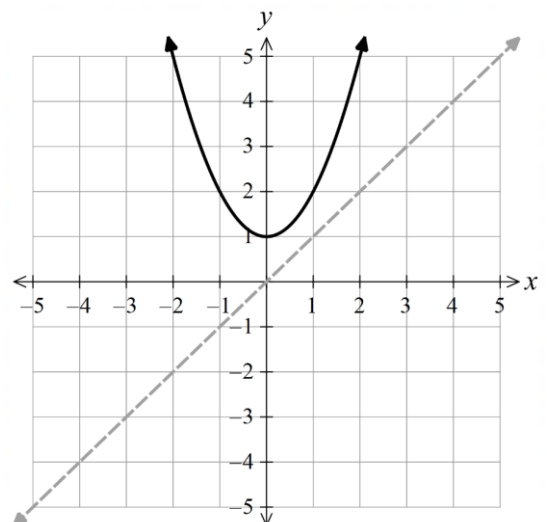
✎ Guided Practice

Reflect these graphs across $y = x$.

a

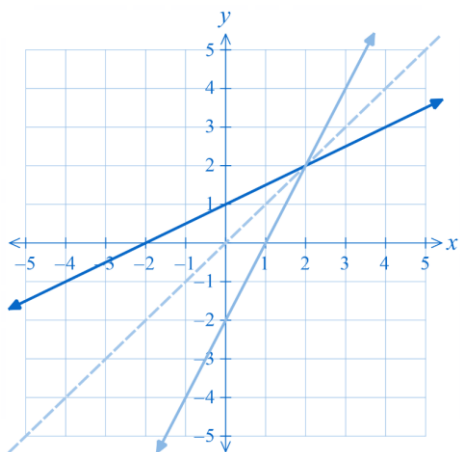


b

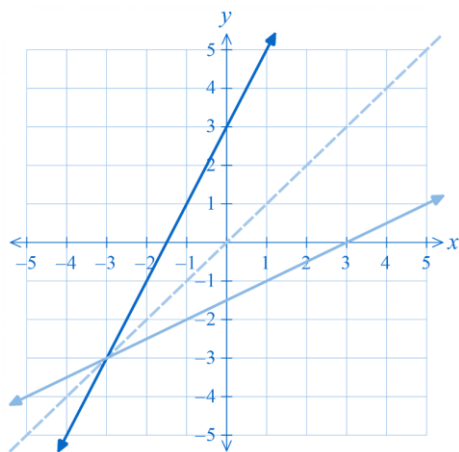


1 Sketch the reflection of these graphs across $y = x$.

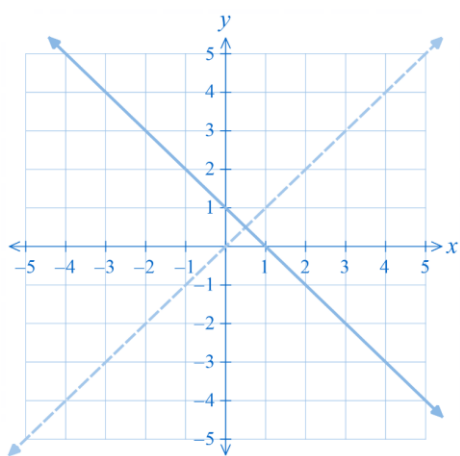
a



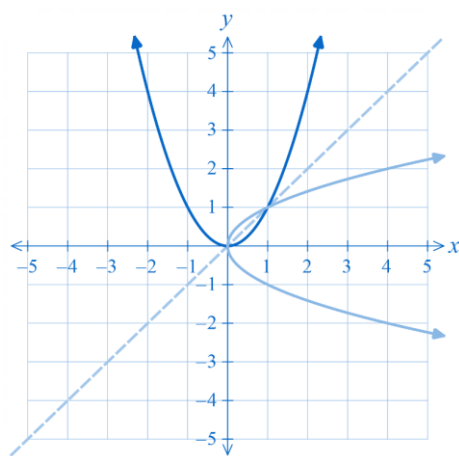
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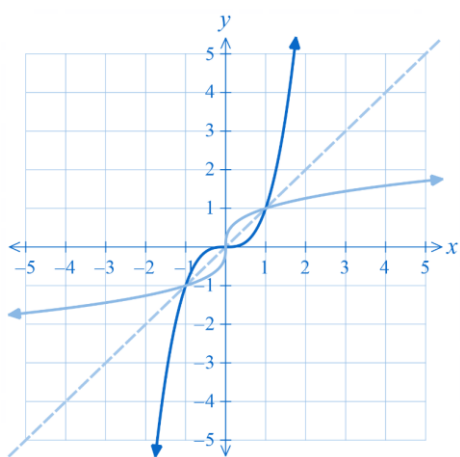
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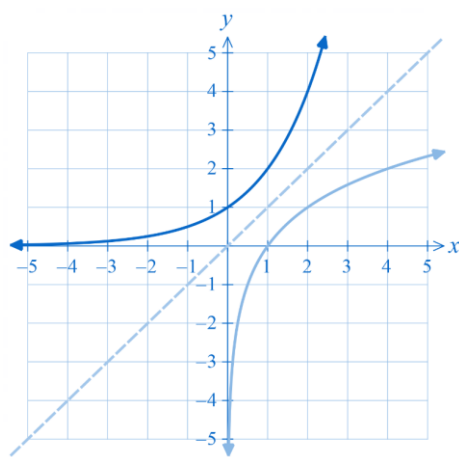
d



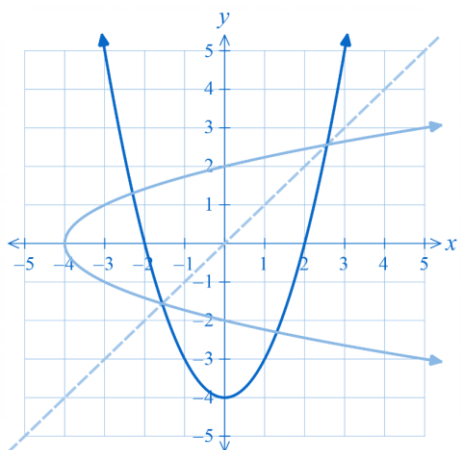
e



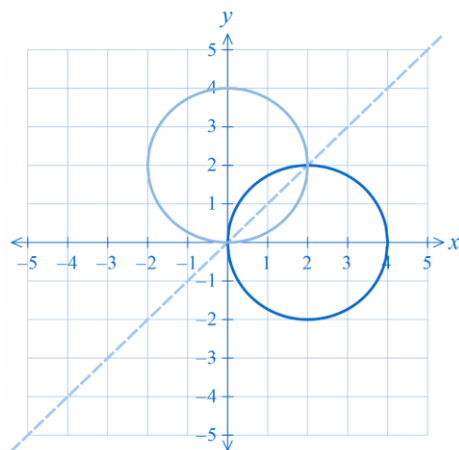
f



g



h



Inverse of a Relation

Q Inverse Relations

Consider a function $f(x)$.

The **inverse** function, $f^{-1}(x)$ will undo the original function.

⚠ $f^{-1}(x)$ is the inverse function and $(f(x))^{-1}$ is the reciprocal function $\frac{1}{f(x)}$.

For a function that doubles all inputs $f(x) = 2x$,

the inverse function is $f^{-1}(x) = \frac{x}{2}$, which halves all inputs.

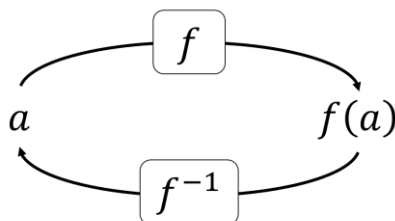
$\sin^{-1}(x)$ is the inverse of $\sin(x)$.

The composition of a function and its inverse is the original input.

$$f^{-1}(f(x)) = f(f^{-1}(x)) = x$$

Example:

Consider the linear function $f(x) = 3x + 5$; the function multiplies the input by 3 and adds 5.



The inverse function would undo this by 5 and by 3.

So, we can write $f^{-1}(x) = \frac{x-5}{3}$

Check that this works using the input $x = 7$

$$f(7) =$$

Then put this into the inverse function.

$$f^{-1}(f(7)) =$$

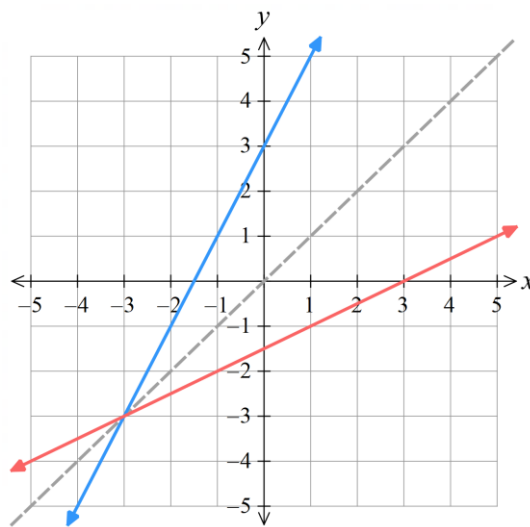
🔍 The Graph of a Relation and its Inverse

“multiply by 2, then add 3”

$$y = 2x + 3$$

“subtract 3, divide by 2”

$$y = \frac{x - 3}{2}$$



What do you notice about the two lines?

For the inverse relation, the coordinates of each pair are

$$\begin{array}{c} (x, y) \\ \downarrow \\ (y, x) \end{array}$$

This is the same as reflecting across the line

This has the effect of **swapping the domain and range**.

📌 More examples: <https://www.desmos.com/calculator/i6pos84gch>

📌 Graph of the Inverse of a Relation

The graph of a relation and its inverse are reflections across the line $y = x$.

The **domain** of the original **becomes the range** of the inverse.

The **range** of the original **becomes the domain** of the inverse.

Equation of the Inverse Relation

- The **inverse** of a function **undoes** what the original function does.
 - The graph of a function and its inverse are **reflections across** $y = x$.
 - Find a function's inverse by **swapping the x and y variables**.
-  The inverse of a function is not always a function. That's why we call it an inverse relation. We will explore this further in a future lesson

Example Determine the inverse relation of a function

Worked Example	Your Turn
Determine the inverse relation of: $y = 5x - 2$ $x = 5y - 2$ $x + 2 = 5y$ $y = \frac{x + 2}{5}$	Determine the inverse relation of: $y = 2x + 1$ $y = \frac{x - 1}{2}$
Determine the inverse relation of: $y = 5x^3$ $x = 5y^3$ $\frac{x}{5} = y^3$ $y = \sqrt[3]{\frac{x}{5}}$	Determine the inverse relation of: $y = x^3 - 2$ $\sqrt[3]{x + 2}$
Determine the inverse relation of: $y = x^2 + 2x$ $x = y^2 + 2y$ $x + 1 = y^2 + 2y + 1$ $x + 1 = (y + 1)^2$ $y + 1 = \pm\sqrt{x + 1}$ $y = -1 \pm \sqrt{x + 1}$	Determine the inverse relation of: $y = x^2 + 4x - 7$ $y = -2 \pm \sqrt{x + 11}$

1 Find the inverse relation.

a $y = 3x$

$y = \frac{x}{3}$

b $y = \sqrt[3]{x}$

$y = x^3$

c $y = \frac{x}{5}$

$y = 5x$

d $y = x - 5$

$y = x + 5$

e $y = x + 3$

$y = x - 3$

f $y = x^3$

$y = \sqrt[3]{x}$

g $y = 3x - 2$

$y = \frac{x+2}{3}$

h $y = \frac{x}{2} + 1$

$y = 2x - 2$

i $y = 3 - \frac{x}{2}$

$y = 6 - 2x$

j $x - y + 1 = 0$

$y = x - 1$

k $2x + 5y - 10 = 0$

$y = -\frac{5x}{2} + 5$

l $y = x^3 + 5$

$y = \sqrt[3]{x-5}$

m $y = \sqrt{x+2}$

$y = x^2 - 2$

n $y = \frac{3}{x+5}$

$y = \frac{3}{x} - 5$

o $y = \frac{3}{x}$

$y = \frac{3}{x}$

2 Find the inverse relation.

a $y = x^2$

$y = \pm\sqrt{x}$

b $y = x^2 + 5$

$y = \pm\sqrt{x-5}$

c $y = 2x^4$

$y = \pm\sqrt[4]{\frac{x}{2}}$

d $y = x^2 + 8x$

$y = \pm\sqrt{x+16} - 4$

e $y = 4x - x^2$

$y = \pm\sqrt{4-x} + 2$

f $y = x^2 + 12x - 11$

$y = \pm\sqrt{x+47} - 6$

- 3 Swap x and y and solve for y to find the inverse of each function.

a

$$y = \frac{1}{x}$$

$$y = \frac{1}{x}$$

b

$$y = -x$$

$$y = -x$$

c

$$y = \frac{2x - 2}{x - 2}$$

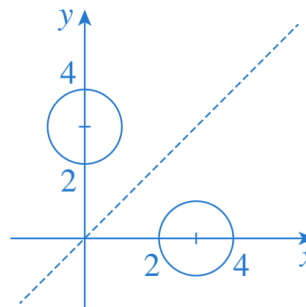
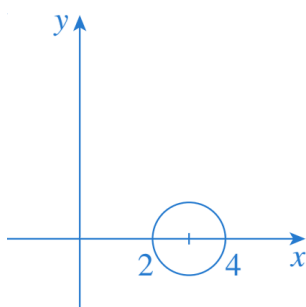
$$y = \frac{2x - 2}{x - 2}$$

What do you notice? What does this mean, graphically?

Each inverse is identical to the original function. The graph is symmetrical about the line $y = x$

- 4 Consider $(x - 3)^2 + y^2 = 1$.

a Graph this relation on the axes below.



b Determine the equation of the inverse relation (do not rearrange to make y the subject)

$$(y - 3)^2 + x^2 = 1$$

c Sketch the relation on the same axes above.

d Write down the domain and range of the original and of the inverse relation.
How are they related?

Original: domain: $2 \leq x \leq 4$, range: $-1 \leq y \leq 1$

Inverse: domain: $-1 \leq x \leq 1$, range: $2 \leq y \leq 4$

The domain and range of the inverse is swapped for a relation's inverse.

5

a State the domain and range of $y = \frac{2}{x-1}$

$$D: x \in \mathbb{R}, x \neq 1$$

$$R: y \in \mathbb{R}, y \neq 0$$

b Find the equation of the inverse.

$$y = \frac{2}{x} + 1$$

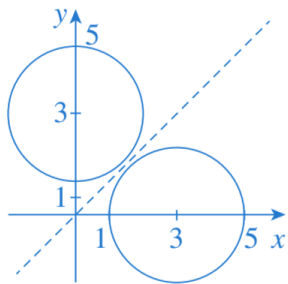
c State the domain and range of the inverse function.

$$D: x \in \mathbb{R}, x \neq 0$$

$$R: y \in \mathbb{R}, y \neq 1$$

6

a Sketch $(x - 3)^2 + y^2 = 9$



State the domain and range.

$$1 \leq x \leq 5$$

$$-3 \leq y \leq 3$$

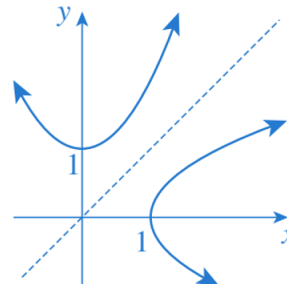
On the same axis, graph the inverse.

State the domain and range of the inverse.

$$-3 \leq x \leq 3$$

$$1 \leq y \leq 5$$

b Sketch $y = x^2 + 1$



State the domain and range.

$$x \in \mathbb{R}$$

$$y \geq 1$$

On the same axis, graph the inverse.

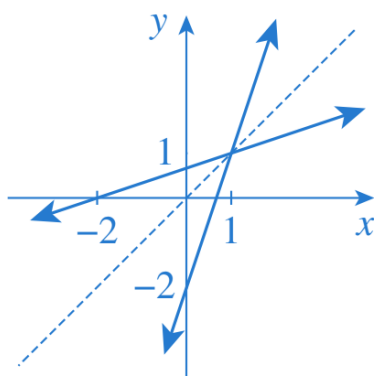
State the domain and range of the inverse.

$$x \geq 1$$

$$y \in \mathbb{R}$$

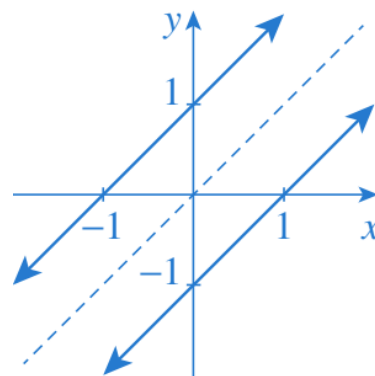
7 Determine each inverse algebraically by swapping x and y and then making y the subject, then sketch the graph of each function and its inverse on the same axis.

a $y = 3x - 2$



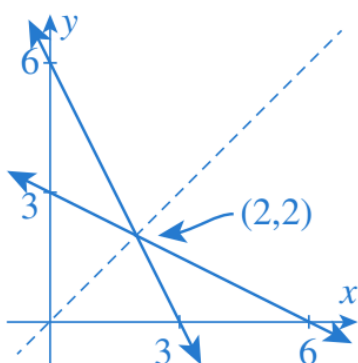
$$y = \frac{x+2}{3}$$

b $y = x + 1$



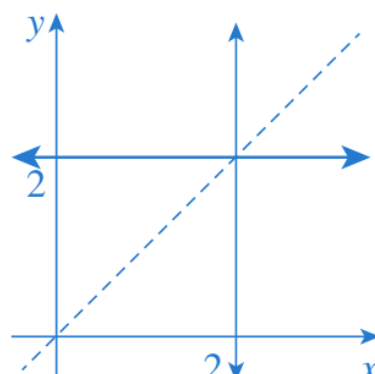
$$y = x - 1$$

c $y = 3 - \frac{1}{2}x$



$$y = 6 - 2x$$

d $y = 2$



$$x = 2$$

- 8 By first identifying the domain and range of these functions, identify the domain and range of the inverse.

a $y = 3x + 4$

D: all real x
R: all real x

b $y = \sqrt{x} - 3$

D: $x \geq -3$
R: $y \geq 0$

c $y = x^2 + 1$, for $x \geq 0$

D: $x \geq 1$
R: $y \geq 0$

d $y = \frac{3}{x}$

D: all real x , $x \neq 0$
R: all real y , $y \neq 0$

e $y = 2^x$

D: $x > 0$
R: all real y

f $y = \frac{1}{\sqrt{x+2}}$

D: $x > 0$
R: $y > -2$

- 9 When there are restrictions on a function, then the x and y must be swapped in these as well.

a Find the inverse of
 $y = 2x - 3$, where $x > 1$
 $x = 2y - 3$, $y > 1$

b Find the inverse of
 $y = 3x - 10$, where $x < 2$

$y = \frac{1}{2}(x + 3)$, $y > 1$

$y = \frac{1}{6}(x + 1)$, $y < 2$

c Find the inverse of
 $y = x^3 + 2$, where $x < 3$

d Find the inverse of
 $y = 13 - 6x$, where $x \geq 3$

$y = \sqrt[3]{x-2}$, $y < 3$

$y = \frac{1}{6}(13 - x)$, $y \geq 3$

- 10 For each part in the question above, your inverse should have a restriction in terms of y . By solving the inequality, rewrite the restriction in terms of x .

a
The inverse was $y = \frac{1}{2}(x + 3)$, $y > 1$
 $\therefore \frac{1}{2}(x + 3) > 1$

$y = \frac{1}{2}(x + 3)$, $x > -1$

b
 $y = \frac{1}{6}(x + 1)$, $x < -4$

c
 $y = \sqrt[3]{x-2}$, $x < 29$

d
 $y = \frac{1}{6}(13 - x)$, $x \leq -5$

- 11 Consider the function $y = \sqrt{2x - 1}$.
- a State the domain and range of this function.

$$\text{Domain: } x \geq \frac{1}{2}$$

$$\text{Range: } y \geq 0$$

- b Find the equation of the inverse and state the domain and range.
You must consider the restrictions on the original function, not just the equation of the inverse.

$$y = \frac{1}{2}(x^2 + 1)$$

$$\text{Domain: } x \geq 0$$

$$\text{Range: } y \geq \frac{1}{2}$$

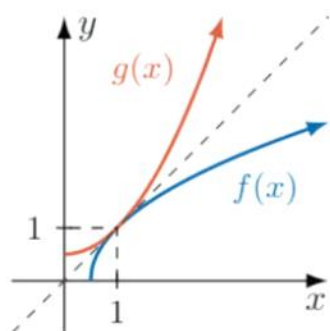
- c Find the coordinates of the point where the original function intersects $y = x$.

(1,1)

- d Explain why this point also represents where the original function intersects with the inverse.

A function and its inverse has symmetry about $y = x$,
so any point on the line $y = x$ will stay the same.

- e Hence, draw $y = \sqrt{2x - 1}$ and its inverse on the same set of axes.



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Given $f(x) = 1 + \sqrt{x}$, what are the domain and range of $f^{-1}(x)$?

- A. $x \geq 0, y \geq 0$
 B. $x \geq 0, y \geq 1$
 C. $x \geq 1, y \geq 0$
 D. $x \geq 1, y \geq 1$

Domain and range of $f(x)$ is $x \geq 0, y \geq 1$

This is swapped for the inverse, $\therefore C$

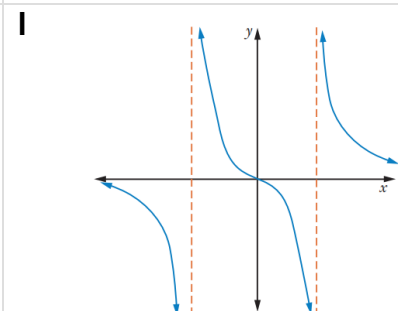
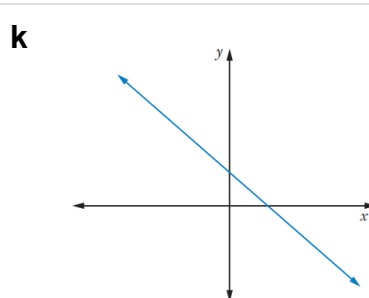
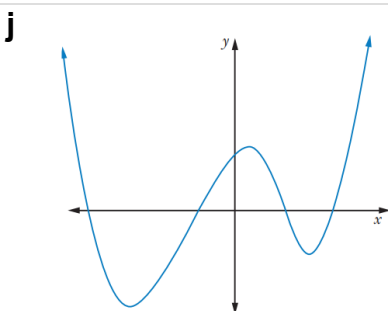
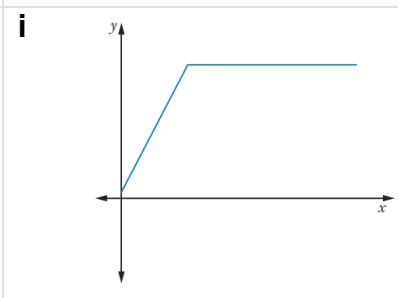
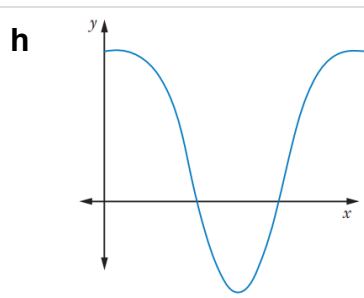
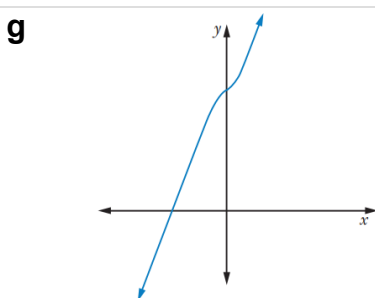
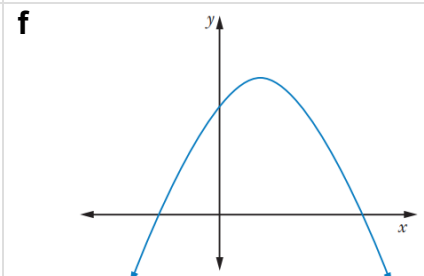
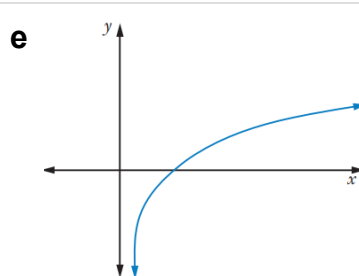
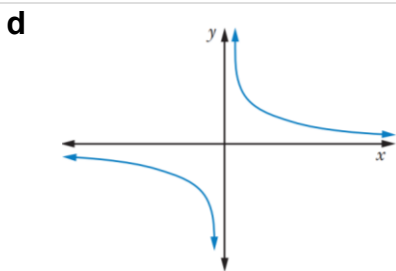
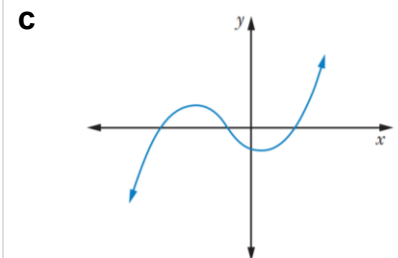
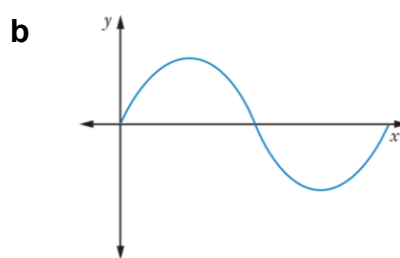
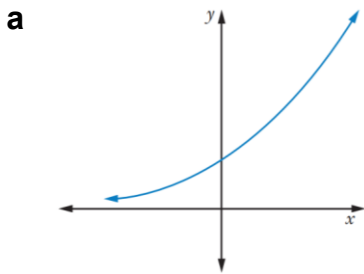
Determining if the Inverse Relation is a Function

i Equation of the Inverse Relation

- The inverse of a function is not always a function itself.
- Remember: if a relation passes the vertical line test, it is a function.
- Since the inverse swaps x and y coordinates, if a function passes the **horizontal line test**, then its **inverse will also be a function**.
- **Even functions do not have an inverse function**, as its symmetry means it will fail the horizontal line test.

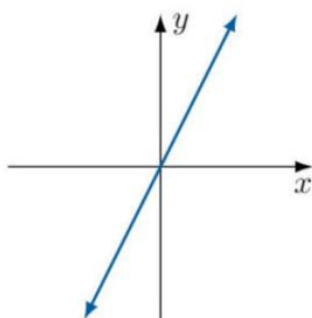
Q Identify relations that will have an inverse function

Which of the following relations will have an inverse function?



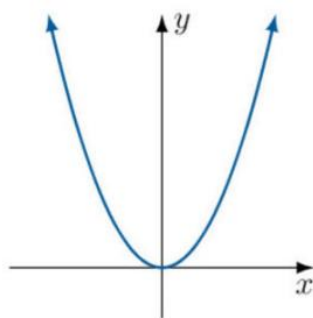
1 For each graph, decide whether the inverse will be a function.

a



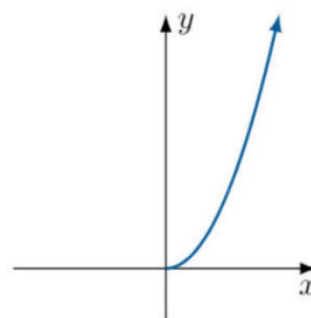
Inverse is a function

b



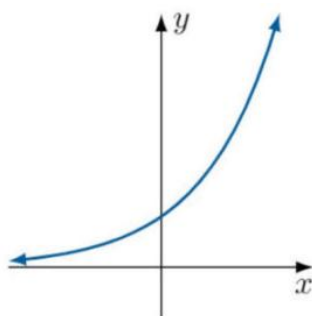
Inverse not a function

c



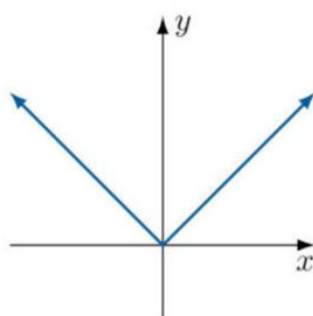
Inverse is a function

d



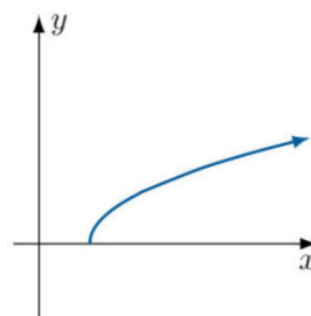
Inverse is a function

e



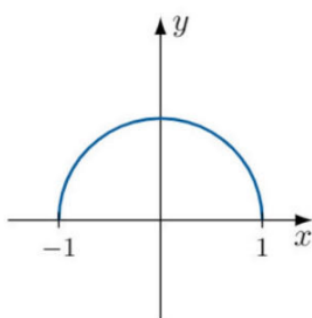
Inverse not a function

f



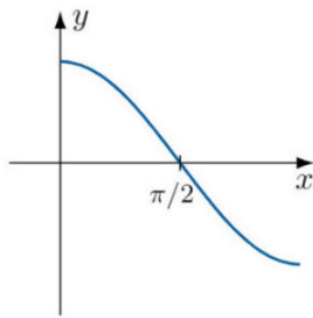
Inverse is a function

g



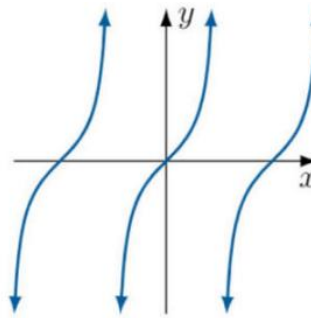
Inverse not a function

h



Inverse is a function

i



Inverse not a function

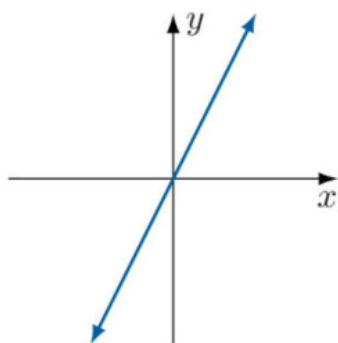
2 Explain why these functions do not have inverse functions.

$$f(x) = x^2 - 5, \quad f(x) = x^4, \quad f(x) = 3^{x^2}, \quad y = |3x - 2|$$

There are all even functions.

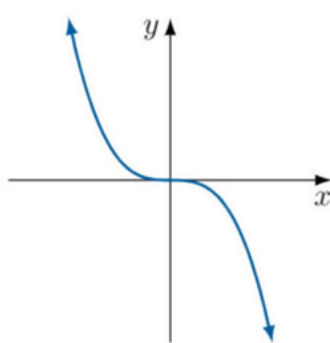
- 3 Classify whether these relations are one-to-one, and therefore whether it is a function that has an inverse functions.

a



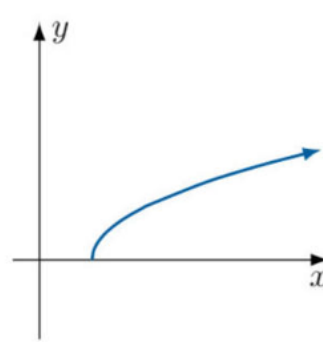
one-to-one

b



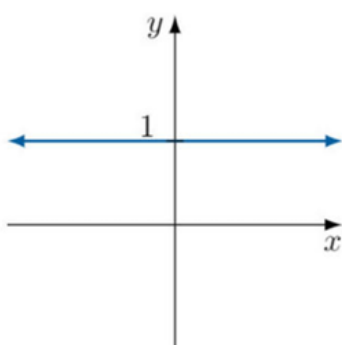
one-to-one

c



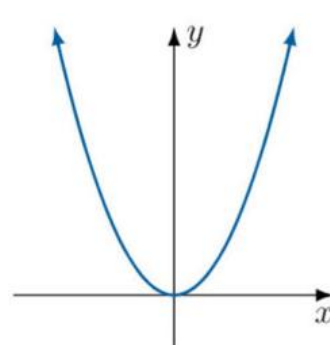
one-to-one

d



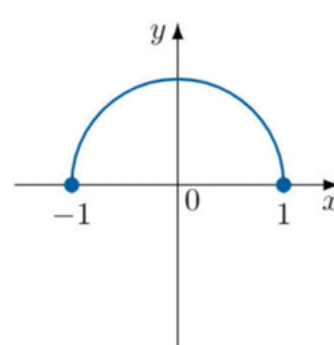
Not one-to-one

e



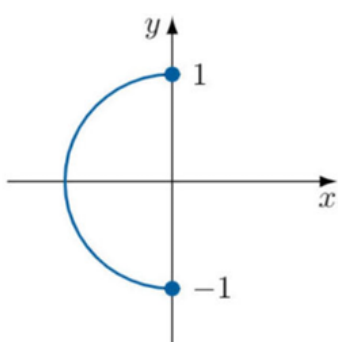
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f



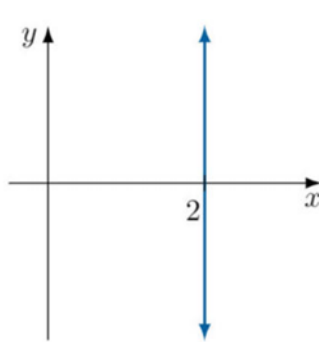
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g



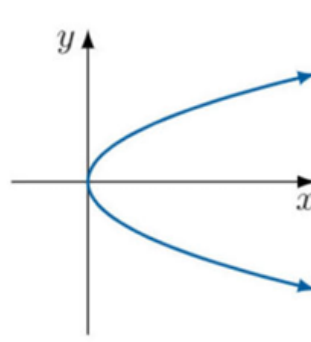
Not one-to-one

h



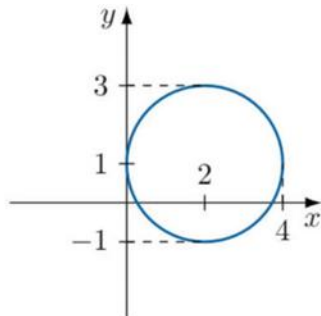
Not one-to-one

i



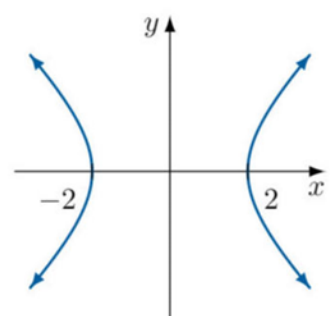
Not one-to-one

j



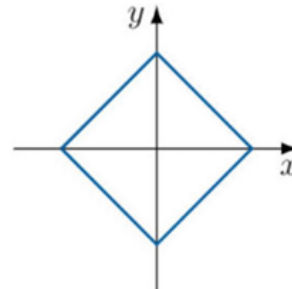
Not one-to-one

k



Not one-to-one

l



Not one-to-one

4 When will the inverse of a relation be a function?

One-to-one and one-to-many relations have an inverse that is a function.

5 When will the inverse of a function be itself a function?

Only one-to-one functions have an inverse that is a function.

6 Explain why the inverse of an even function will not be a function.

Even functions satisfy $f(-x) = f(x)$, they have at least 2 x -values for each y -value.
Therefore, the inverse will have at least 2 y -values for each x -value, so is not a function.

7 Emmy says:



The inverse of an odd function will always be a function.
An example is $y = 2x$.

Show that she is incorrect by giving a counterexample.

A counterexample is $y = x(x - 1)(x + 1)$. This is an odd function that fails the horizontal line test.

8 Show that $f(x) = \frac{2^x + 2^{-x}}{2}$ does not have an inverse function.

$$f(-x) = \frac{2^{-x} + 2^x}{2} = f(x)$$

This is an even function, so its inverse is not a function.

Inverse Functions

① Finding the Inverse Function

The inverse function is the set of original ordered pairs, but with **input and output swapped**.

$$f = \{(a, A), (b, B), (c, C)\}$$

$$f^{-1} = \{(A, a), (B, b), (C, c)\}$$

If you are given $f(x)$ and asked to find $f^{-1}(x)$:

1. Rewrite $f(x) =$ as $y =$
2. Swap the x and y .
3. Rearrange for y , then write as $f^{-1}(x)$.

 **Example** Find $f^{-1}(x)$ given $f(x)$, no domain restrictions

$f(x) = x^3 + 2$, Find $f^{-1}(x)$.

1. $y = x^3 + 2$

2. $x = y^3 + 2$

3. $y^3 = x - 2$

$$y = \sqrt[3]{x - 2}$$

$$f^{-1}(x) = \sqrt[3]{x - 2}$$

Guided Practice

Find the inverse function of each of these functions.

a $f(x) = 3x - 8$

$$f^{-1}(x) = \frac{x + 8}{3}$$

b $f(x) = 2x^5 + 7$

$$f^{-1}(x) = \sqrt[5]{\frac{x - 7}{2}}$$

c $f(x) = \frac{1}{x - 2}$

$$f^{-1}(x) = \frac{1}{x} + 2$$

① Solving Problems involving Inverses

Many problems can be solved by applying the function to both sides of the equation.
This helps us avoid finding the inverse function.

🔧 Example Solve problems involving inverse functions

$h(x) = 2^{5x}$. Find the value of x when $h^{-1}(x) = 3$

$$h^{-1}(x) = 3$$

$$h(h^{-1}(x)) = h(3)$$

$$x = h(3)$$

$$x = 2^{15}$$

✎ Guided Practice

a $j(x) = 3^x$. Find x when $j^{-1}(x) = 4$.

$$x = 81$$

b $g(x) = \frac{4x+1}{x-2}$. Find the value of x when $g^{-1}(x) = 5$.

$$x = 7$$

c $f(x) = 2x^3 - 7$. Find the value of x when $f^{-1}(x) = -2$.

$$x = -23$$

d $h(x) = 5^{x+1}$. Find $h^{-1}(125)$.

Hint: let $y = h^{-1}(125)$ and apply h to both sides, then solve.

$$x = 2$$

9 Find the inverse function of:

a $f(x) = 2x + 5$

b $f(x) = 4 - 3x$

$$f^{-1}(x) = \frac{x-5}{2}$$

$$f^{-1}(x) = \frac{4-x}{3}$$

c $f(x) = x^3 - 2$

d $f(x) = \frac{1}{x-5}$

$$f^{-1}(x) = \sqrt[3]{x+2}$$

$$f^{-1}(x) = \frac{1}{x} + 5$$

e $f(x) = \frac{x+1}{6}$

f $f(x) = 12 - 3x$

$$f^{-1}(x) = 6x - 1$$

$$f^{-1}(x) = 4 - \frac{x}{3}$$

g $f(x) = \frac{4}{x+2}$

h $f(x) = \frac{5}{x+3}$

$$f^{-1}(x) = \frac{4-2x}{x}$$

$$f^{-1}(x) = \frac{5}{x} - 3$$

10 Let $f(x) = 2x - 8$ and $g(x) = \frac{1}{2}x + 4$.

a Find:

i $g(f(5))$

ii $f(g(5))$

5

5

b Find:

i $g(f(x))$

ii $f(g(x))$

x

x

c What can you conclude about $f(x)$ and $g(x)$?

$g(x)$ is the inverse function of $f(x)$ (and vice versa)

- 11 Given $g(x) = 3x^3 - 1$
a Find an expression for $g^{-1}(x)$

$$g^{-1}(x) = \sqrt[3]{\frac{x+1}{3}}$$

- b Solve the equation $g^{-1}(x) = 2$

$$x = 23$$

- 12 Find the inverse of each of the functions $g(x) = 6 - x$ and $h(x) = \frac{3}{x}$
What do you notice?

$$g^{-1}(x) = 6 - x, \quad h^{-1}(x) = \frac{3}{x}$$

The inverse function is the original function (involution)

- 13 Let f be the function that converts a temperature in Celsius, x , to Fahrenheit.

$$f(x) = \frac{9}{5}x + 32$$

Find the inverse function that converts Fahrenheit to Celsius.

$$f^{-1}(x) = \frac{5}{9}(x - 32)$$

Where x is temperature in Fahrenheit

- 14 Given $h(x) = 3x + 19$ and $k(x) = 2x - 10$, work out the value of x for which $h^{-1}(x) = k^{-1}(x)$

$$x = -68$$

- 15 Find the inverse of $h(x) = \frac{3x+4}{5x-6}$

$$h^{-1}(x) = \frac{6x+4}{5x-3}$$

16 For each function, find its inverse $f^{-1}(x)$ then verify $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$.

a $f(x) = x^3 + 1$

Verify $f(f^{-1}(x)) = x$ $f^{-1}(x) = \sqrt[3]{x-1}$

Verify $f^{-1}(f(x)) = x$

b $f(x) = \frac{1-x}{3+x}$

Verify $f(f^{-1}(x)) = x$ $f^{-1}(x) = \frac{1-3x}{1+x}$

Verify $f^{-1}(f(x)) = x$

c $f(x) = \sqrt[3]{x-7}$

Verify $f(f^{-1}(x)) = x$ $f^{-1}(x) = x^3 + 7$

Verify $f^{-1}(f(x)) = x$

17 $f(x) = 3x + 4$ and $g(x) = ax^2$

a Work out an expression for $f^{-1}(x)$

b Given that $f(g(3)) = 85$, find the value of a .

$$f^{-1}(x) = \frac{x-4}{3}$$

$$g(3) = 9a$$

$$f(9a) = 27a + 4 = 85$$

$$a = 3$$

- 18** $f(x) = ax + 8$ and $g(x) = 2x - 3$
- a** Work out an expression for $f^{-1}(x)$
- b** Given $fg(2) = 13$, find the value of a .

$$\begin{aligned} f^{-1}(x) &= \frac{x-8}{a} \\ g(2) &= 1 \\ f(1) &= a+8=13 \\ a &= 5 \end{aligned}$$

- 19** $f(x) = x^2 - 1$ and $g(x) = 4x + 5$
- a** Work out an expression for $f^{-1}(x)$
- b** Work out an expression for $g^{-1}(x)$
- c** Given that $fg(x) = 3gf(x)$, show that $4x^2 + 40x + 21 = 0$

$$f^{-1}(x) = \sqrt{x+1}, \quad g^{-1}(x) = \frac{x-5}{4}$$

$$fg(x) = (4x+5)^2 - 1 = 16x^2 + 40x + 24$$

$$gf(x) = 4(x^2 - 1) + 5 = 4x^2 + 1$$

$$\begin{aligned} 16x^2 + 40x + 24 &= 3(4x^2 + 1) \\ 4x^2 + 40x + 21 &= 0 \end{aligned}$$

- 20** $f(x) = 2x + 4$ and $g(x) = 3x - 8$
- a** Find $f^{-1}(x)$ and $g^{-1}(x)$
- b** Find an expression for $f(g(x))$
- c** Find an expression for the inverse of $f(g(x))$
- d** Is $(fg)^{-1}(x)$ the same as $f^{-1}(g^{-1}(x))$?
- e** Is $(fg)^{-1}(x)$ the same as $g^{-1}(f^{-1}(x))$?

$$\begin{aligned} f^{-1}(x) &= \frac{x-4}{2}, \quad g^{-1}(x) = \frac{x+8}{3} \\ fg(x) &= 2(3x-8) + 4 = 6x - 12 \\ (fg)^{-1}(x) &= \frac{x+12}{6} \\ f^{-1}g^{-1}(x) &= \frac{x-4}{6} \quad \text{not the same as } (fg)^{-1}(x) \\ g^{-1}f^{-1}(x) &= \frac{x+12}{6} \quad \text{yes, the same as } (fg)^{-1}(x) \end{aligned}$$

21

- a Let $f(x)$ be the function “increase x by 50%”. Write the equation for $f(x)$.

$$f(x) = 1.5x$$

- b A student in Year 8 says:



If you increase by 50%, then you can get back to the original by decreasing by 50%.

Let $g(x)$ be “decrease x by 50%.”

- i Write the equation for $g(x)$.

$$g(x) = 0.5x$$

- ii Show that she is incorrect by finding $g(f(x))$.

$$g(1.5x) = 0.5(1.5x) = 0.75x$$

Since $g(f(x)) \neq x$, $g(x)$ is not the inverse of $f(x)$

Therefore, a decrease of 50% will not undo an increase of 50%

- c What is the inverse function $f^{-1}(x)$?

$$f^{-1}(x) = \frac{x}{1.5} = \frac{2x}{3}$$

- d Therefore, to undo a 50% increase, what percentage do you need to decrease by?

$$f^{-1}(x) = \frac{2}{3}x, \text{ we need to decrease } x \text{ by a third,}$$

so 33.3% decrease

22

- a Find the inverse function of $f(x) = \frac{1-x}{1+x}$.

$$f^{-1}(x) = \frac{1-x}{1+x}$$

- b What does this tell you about $f(f(x))$?

$f(x)$ and $f^{-1}(x)$ is the same. So, if $f(x)$ is applied twice, you get the original back.

- c What does this tell you about the graph of $f(x)$?

It is symmetrical about the line $y = x$

23

- a** Show that $\frac{x^2+x}{x^2+4x+3}$ can be written as $\frac{x}{x+a}$ where a is a positive integer.

$$\frac{x(x+1)}{(x+1)(x+3)} = \frac{x}{x+3}, \quad a = 3$$

- b** Hence, given $f(x) = \frac{x^2+x}{x^2+4x+3}$, find an expression for $f^{-1}(x)$

$$f^{-1}(x) = \frac{3x}{1-x}$$

- 24** $h(x) = \frac{2x+1}{x-1}$. Find an expression for $h^{-1}(x)$

$$h^{-1}(x) = \frac{x+1}{x-2}$$

25

- a** Find an expression for $f^{-1}(x)$ when $f(x) = \frac{3x+7}{2x-8}$

$$f^{-1}(x) = \frac{8x+7}{2x-3}$$

- b** Find a general expression for $g^{-1}(x)$ when $g(x) = \frac{ax+b}{cx+d}$

$$g^{-1}(x) = \frac{dx-b}{a-cx}$$

26 $f(x) = x + 2$ and $g(x) = \frac{1}{x-1}$

- a** Find $g^{-1}(x)$
- b** Find $f(g(x))$ and give your answer in its simplest form.
- c** Find the inverse of $f(g(x))$, i.e. $(fg)^{-1}(x)$

$$\begin{aligned} g^{-1}(x) &= \frac{1}{x} + 1 \\ fg(x) &= \frac{1}{x-1} + 2 = \frac{2x-1}{x-1} \\ (fg)^{-1}(x) &= \frac{x-1}{x-2} \end{aligned}$$

27 $f(x) = 4x + 2$ and $g(x) = px + q$.

If $g(4) = 7$ and $f^{-1}(18) = g(3)$, work out the values of p and q .

$$p = 3, \quad q = -5$$

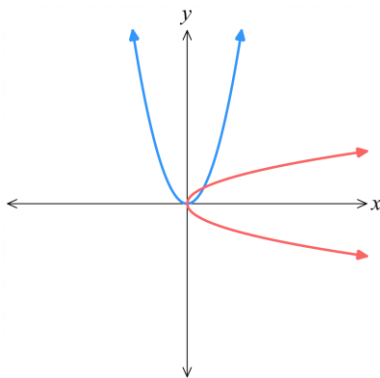
Domain Restrictions

i Restricting the Domain

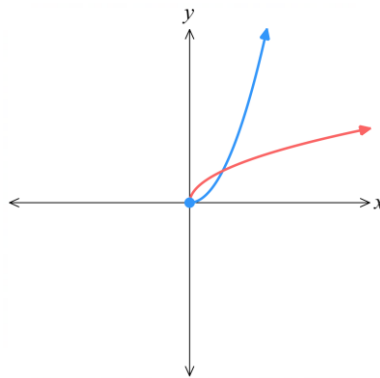
When a function's inverse is not a function, we need to restrict the domain so that the new restricted function has an inverse.

- We call a function **invertible** if its inverse is also a function.
- A function is invertible when it is **one-to-one**.
- Make a function invertible by limiting its domain based on **turning points**.
- Restrict domains so that the curve is **strictly increasing** or **strictly decreasing**, i.e., the gradient does not change sign, and include the turning point.

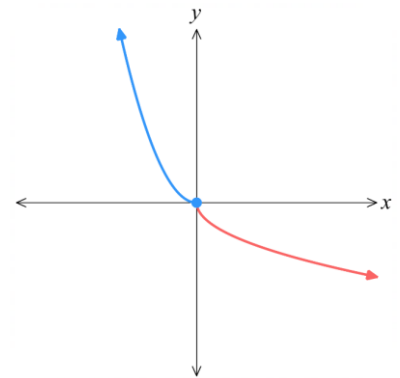
Consider $y = x^2$, inverse is $y^2 = x$ (or $y = \pm\sqrt{x}$), which is not a function.



If we restrict $y = x^2$ to $x \geq 0$, inverse is $y = \sqrt{x}, y \geq 0$, which is a function.



If we restrict $y = x^2$ to $x \leq 0$, inverse is $y = -\sqrt{x}, y \leq 0$, which is a function.



i Finding the Inverse Function (Domain Restrictions)

If the original function has a restriction, find the domain of the inverse by:

- Using the range of the original function as the domain of the inverse, or
- Sketching the reflection of the original function, or
- Solving the relevant inequality

Example Find $f^{-1}(x)$ given $f(x)$ and a domain restriction

$f(x) = 6 - 2x, x > 0$. Find $f^{-1}(x)$ and the new domain.

1. $y = 6 - 2x, x > 0$

2. $x = 6 - 2y, y > 0$

3. $-2y = x - 6 \quad y > 0$

$$y = -\frac{x}{2} + 3 \quad y > 0$$

$$f^{-1}(x) = -\frac{x}{2} + 3 \quad y > 0$$

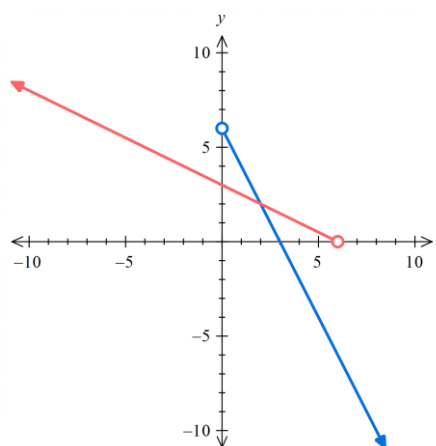
New domain: $-\frac{x}{2} + 3 > 0$

$$-\frac{x}{2} > -3$$

$$x < 6$$

(we can also consider the original range)

$$\therefore f^{-1}(x) = -\frac{x}{2} + 3 \quad x < 6$$



Guided Practice

Find the inverse function and any domain restrictions

a $f(x) = 2x - 5, x > 3$

$$f^{-1}(x) = \frac{x+5}{2}, x > 1$$

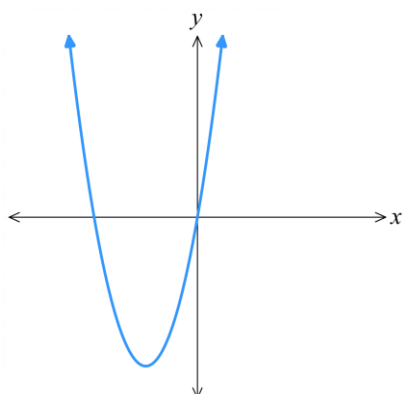
b $f(x) = \sqrt{x} + 5$

Hint: to find the new domain, the easiest method is to consider the original range.

$$f^{-1}(x) = (x-5)^2, x \geq 5$$

1

- a** The graph of $y = x(x + 6)$ is shown. Find the coordinates of its turning point.



- b** Find the domain over which the function $y = x(x + 6)$ is one-to-one and increasing, including the turning point.

$(-3, -9)$

$[-3, \infty)$

- c** Find the inverse over this restricted domain, stating its domain and range.

$$y = \sqrt{x + 9} - 3$$

$$D: [-9, \infty)$$

$$R: [-3, \infty)$$

- d** Find the domain over which the function $y = x(x + 6)$ is one-to-one and decreasing, including the turning point.

$(-\infty, -3]$

- e** Find the inverse over this restricted domain, stating its domain and range.

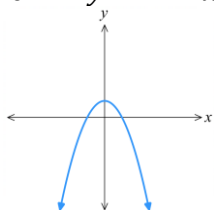
$$y = -\sqrt{x + 9} - 3$$

$$D: [-9, \infty)$$

$$R: (-\infty, -3]$$

- 2** For each quadratic function, use the turning point to write down two ways to restrict the domain so that the inverse function exists.

a $y = 1 - x^2$



b $y = (x - 2)^2$

$x \geq 0$ and $x \leq 0$

$x \geq 2$ and $x \leq 2$

c $y = (x - 3)(x + 1)$

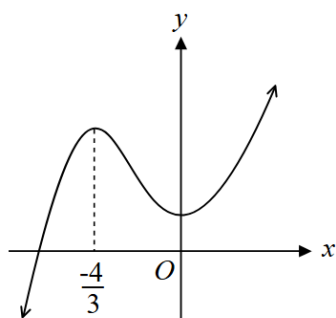
d $y = x^2 - 6x + 8$

$x \geq 1$ and $x \leq 1$

$x \geq 3$ and $x \leq 3$

3 2019 VCE Methods Band 4

The function $f(x) = 5x^3 + 10x^2 + 1$ is shown below.
In what domain will it have an inverse function?



- A. $D = (-2, \infty)$
- B. $D = \left(-\infty, \frac{1}{2}\right]$
- C. $D = (-\infty, -1]$
- D. $D = [0, \infty)$

D

4 $f(x) = (x - 1)^2 - 3$

a Find the largest domain containing $x = 3$ such that the inverse is defined.

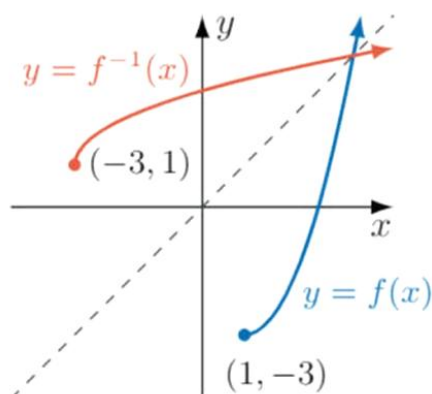
$x \geq 1$

b State the domain and range of $f^{-1}(x)$.

D: $x \geq -3$

R: $y \geq 1$

c Sketch the graph of $y = f(x)$ with the restricted domain and $y = f^{-1}(x)$ on the same axes.



5 Find the inverse function $f^{-1}(x)$, specifying the domain of the inverse. A sketch may help.

a $f(x) = 9 - x^2, x \geq 0$

b $f(x) = x^2, x \leq 0$

c $f(x) = x^2 - 2x, x \geq 1$

$f^{-1}(x) = \sqrt{9 - x}, x \geq 9$

$f^{-1}(x) = -\sqrt{x}, x \geq 0$

$f^{-1}(x) = 1 + \sqrt{1 + x}, x \geq 1$

6 Consider $f(x) = x^2 - 2x$

a Find the largest domain containing $x = 2$ such that the inverse is a function.

Turning point at $x = 1$.

$$x \geq 1$$

b State the range of $f(x)$ in that domain.

At $x = 1, y = -1$

$$y \geq -1$$

c Show that the inverse relation is $y^2 - 2y - x = 0$.

$$x = y^2 - 2y$$

$$y^2 - 2y - x = 0$$

d By quadratic formula or completing the square, show that making y the subject yields

$$y = 1 \pm \sqrt{1+x}$$

$$y = \frac{2 \pm \sqrt{4 - 4(1)(-x)}}{2}$$

$$y = \frac{2 \pm \sqrt{4+4x}}{2}$$

$$y = \frac{2 \pm 2\sqrt{1+x}}{2}$$

$$y = 1 \pm \sqrt{1+x}$$

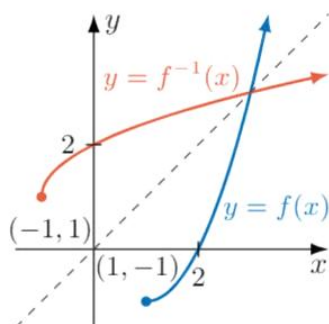
e State the range of the inverse function.

$$y \geq 1$$

f Hence, determine whether the inverse function is $y = 1 + \sqrt{1+x}$ or $y = 1 - \sqrt{1+x}$

$$y = 1 + \sqrt{1+x}$$

g Hence sketch the graph of $y = f(x)$ and $y = f^{-1}(x)$ on the same axes.



7 2012 HSC Extension 1 Band 4

Let $f(x) = \sqrt{4x - 3}$

- a Find the domain and range of $f(x)$.

$$4x - 3 \geq 0$$

$$x \geq \frac{3}{4}$$

$$y \geq 0$$

- b Find an expression for $f^{-1}(x)$, stating the restriction.

$$f^{-1}(x) = \frac{1}{4}x^2 + \frac{3}{4}, \quad x \geq 0$$

- c Find the points where $y = f(x)$ and $y = x$ intersect.

$$x = \sqrt{4x - 3}$$

$$x^2 = 4x - 3$$

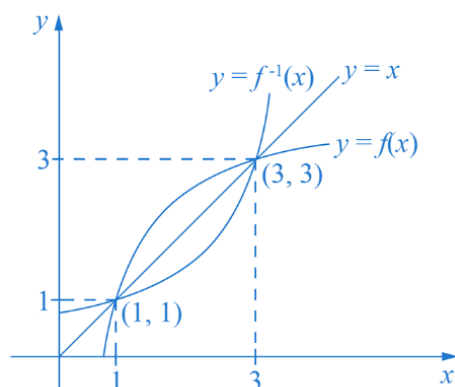
$$x^2 - 4x + 3 = 0$$

$$(x - 3)(x - 1) = 0$$

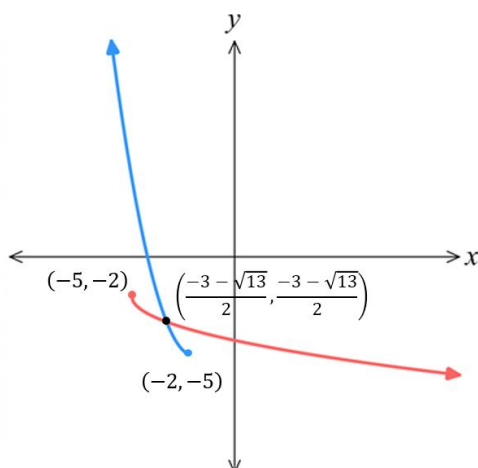
$$x = 1, 3$$

$$\therefore (1, 1), (3, 3)$$

- d Therefore, on the same set of axes, sketch $y = f(x)$ and $y = f^{-1}(x)$.



- 8 Let $f(x) = (x + 2)^2 - 5$. Restrict $f(x)$ to the domain in which it is strictly decreasing and sketch the restricted function and its inverse $f^{-1}(x)$ on the same axis, clearly showing the point(s) of intersection.



$$f(x) = (x + 2)^2 - 5, \quad x \leq -2$$

$$f^{-1}(x) = -\sqrt{x + 5} - 2$$

Point of intersection:

$$\text{Let } f(x) = x$$

$$(x + 2)^2 - 5 = x$$

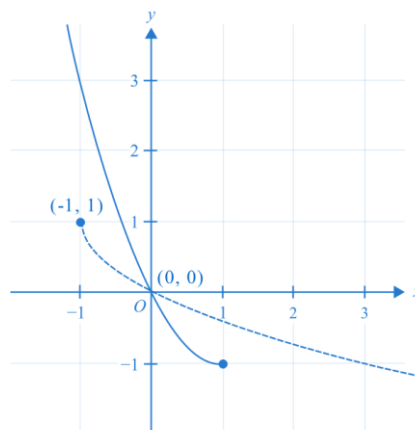
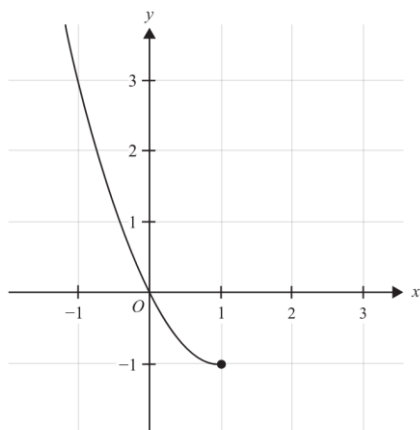
$$\therefore x^2 + 3x - 1 = 0$$

$$\text{Using quadratic formula, } x = \frac{-3 \pm \sqrt{13}}{2}$$

$$\text{Since } x \leq -2, \text{ point of intersection at } \left(\frac{-3 - \sqrt{13}}{2}, \frac{-3 - \sqrt{13}}{2} \right)$$

9 2023 VCE Methods Band 5

$f(x) = x^2 - 2x$ is given below, with domain restriction $x \in (-\infty, 1]$.



a State the range of f .

$[-1, \infty)$

b Sketch the graph of the inverse function on the axes above. Label any endpoints and intercepts.

c Determine the equation of the inverse function $f^{-1}(x)$ and its domain.

$$x = y^2 - 2y$$

$$\text{Complete the square } x = (y - 1)^2 - 1$$

$$x + 1 = (y - 1)^2$$

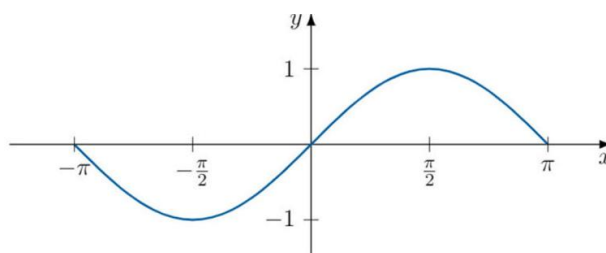
$$y - 1 = \pm\sqrt{x + 1}$$

By inspecting the sketch, consider the negative only

$$f^{-1}(x) = 1 - \sqrt{x + 1}$$

$$D: [-1, \infty)$$

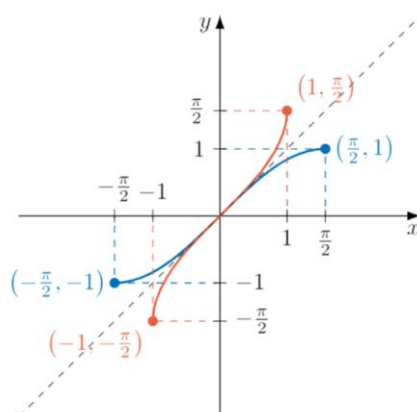
10 The diagram shows $f(x) = \sin x$ in the domain $x \in [-\pi, \pi]$.



a Find the largest possible domain such the inverse function exists.

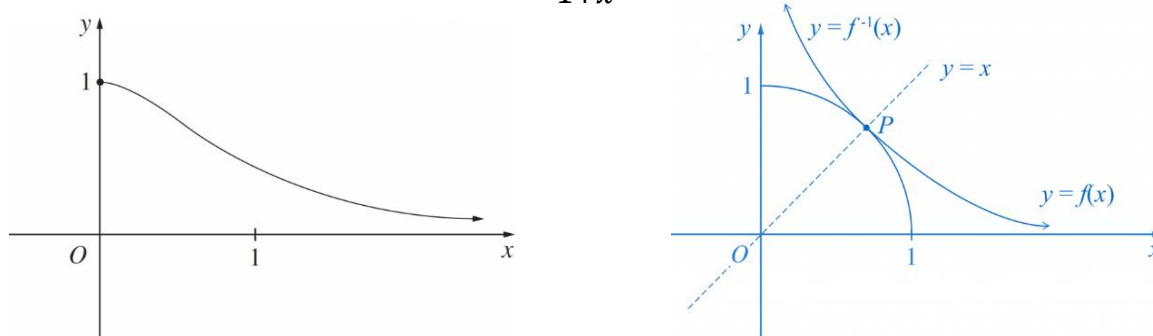
$$-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

b Sketch $y = f(x)$ with the restricted domain and $f^{-1}(x)$ on the same axis.



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The diagram below shows $y = f(x) = \frac{1}{1+x^2}$ for $x \geq 0$.



a On the same set of axes, sketch $y = f^{-1}(x)$.

b State the domain of $y = f^{-1}(x)$.

(0,1]

c Find an expression for $y = f^{-1}(x)$.

$$y = \sqrt{\frac{1-x}{x}}, \text{ as } y \geq 0$$

d $y = f(x)$ and $y = f^{-1}(x)$ meet at a point P , let α be the x -coordinate of P . Show that α is a root of the equation $x^3 + x - 1 = 0$

P occurs where $y = f(x)$ intersects with $y = x$

$$\frac{1}{1+x^2} = x$$

$$1 = x(1+x^2)$$

$$1 = x + x^3$$

$$x^3 + x - 1 = 0$$

$\therefore$ The root of $x^3 + x - 1 = 0$ will be the x -coordinate of P .

12 Let $f(x) = x + \frac{1}{x}$

a Show that $f\left(\frac{1}{a}\right) = f(a)$.

$$f\left(\frac{1}{a}\right) = \frac{1}{a} + \frac{1}{\left(\frac{1}{a}\right)}$$

$$= \frac{1}{a} + a$$

$$= f(a)$$

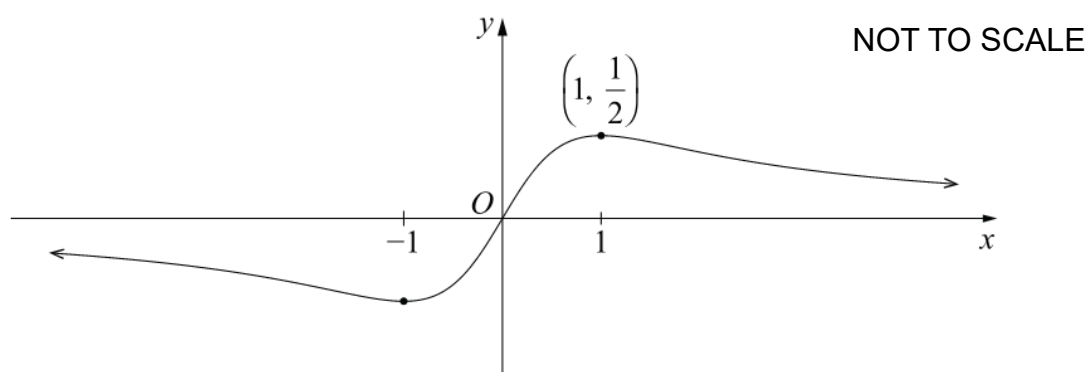
b Hence explain why $f(x)$ does not have an inverse function.

Since $f(a)$ and $f\left(\frac{1}{a}\right)$ are equal, there are two x values that produce the same y -value.

This means the function is not one-to-one, so there won't be an inverse function.

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The diagram shows $y = \frac{x}{x^2+1}$



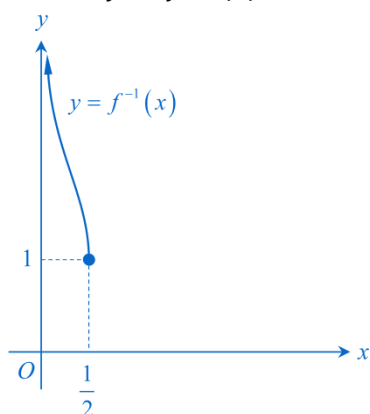
Consider $f(x) = \frac{x}{x^2+1}$ for $x \geq 1$

a State the domain and range of $f^{-1}(x)$.

Domain and range of $f(x)$: D: $x \geq 1$ R: $y \leq \frac{1}{2}$

$\therefore$ Domain and range of $f^{-1}(x)$: D: $x \leq \frac{1}{2}$ R: $y \geq 1$

b Sketch $y = f^{-1}(x)$ on a separate axis below.



c Find an expression for $f^{-1}(x)$.

$$x = \frac{y}{y^2+1}$$

$$xy^2 + x = y$$

$$xy^2 - y + x = 0$$

$$y = \frac{1 \pm \sqrt{1-4x^2}}{2x}$$

Since $y \geq 1$

$$f^{-1}(x) = \frac{1 + \sqrt{1-4x^2}}{2x}$$

d Prove algebraically that $f(x)$ and $f^{-1}(x)$ do not intersect.

Let $f(x) = x$

$$\frac{x}{x^2+1} = x$$

$$x = x^3 + x$$

$$x^3 = 0$$

$$x = 0$$

The two graphs will intersect at $x = 0$, however, $f(x)$ has a restriction $x \geq 1$, so there are no points of intersection.

- 14** Decide whether the inverse of $y = \frac{2^x - 2^{-x}}{2}$ is a function.
(You will need to differentiate it to decide whether it is strictly increasing or decreasing).

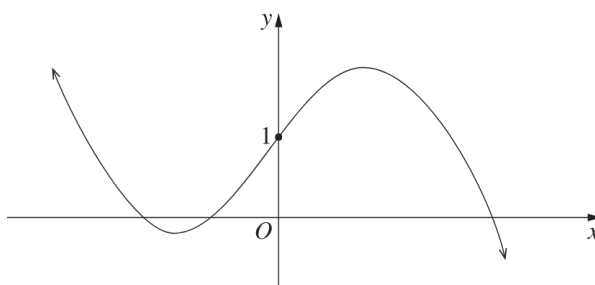
$$f'(x) = \frac{\ln 2(2^x + 2^{-x})}{2}$$

$f'(x)$ is always positive so it is strictly increasing in its entire domain.

This means the inverse will be a function.

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The graph of a cubic function $y = f(x)$ is shown below.



Which of these functions has an inverse relation whose graph has more than 3 points with an x -coordinate of 1?

- A.** $y = \frac{1}{f(x)}$
- B.** $y = f(|x|)$
- C.** $y = |f(x)|$

Draw a quick sketch of each graph.

The one with more than 3 points of intersection with the line $y = 1$ will have an inverse with more than 3 points with x -coordinate 1. $\therefore D$

Draw a quick sketch of each graph.

The one with more than 3 points of intersection with the line $y = 1$ will have an inverse with more than 3 points with x -coordinate 1.

$\therefore D$